HANDS-ON 4 - WEB SERVER LOG ANALYSIS (REAL APACHE ACCESS LOG)

Task	Find a free sample Apache `access.log` online, open it, and count how many requests came from the top 3 IP addresses
Source of the log	Public sample "Apache Logs" dataset from the elastic/examples repository on GitHub (Common Data Formats)
File used	`apache_logs` - 10,000 lines, 2,370,789 bytes, Apache combined log format
Coverage	17 May 2015 10:05 to 20 May 2015 21:05 UTC (all timestamps in the log are +0000)
Method	Opened in a text editor for inspection, then counted with the classic `awk` / `cut` / `sort` / `uniq` pipeline and cross-checked in a spreadsheet

1. What the file looks like

```
83.149.9.216 - - [17/May/2015:10:05:03 +0000] "GET /presentations/logstash-monitorama-2013/images/kibana-search.png HTTP/1.1" 200 203023 "http://semicomplete.com/presentations/logstash-monitorama-2013/" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_9_1) AppleWebKit/537.36 ..."
```

The combined format has seven fields, which map directly onto the questions an analyst asks:

Field	Value in the example	Question it answers
Client IP	`83.149.9.216`	Who
RFC 931 identity	`-`	(unused)
Authenticated user	`-`	Which account
Timestamp	`[17/May/2015:10:05:03 +0000]`	When
Request line	`"GET /path HTTP/1.1"`	What
Status code	`200`	Result
Response size	`203023`	How much data
Referer	`"http://semicomplete.com/..."`	Where they came from
User-Agent	`"Mozilla/5.0 (Macintosh...)"`	With what tool

2. The headline numbers

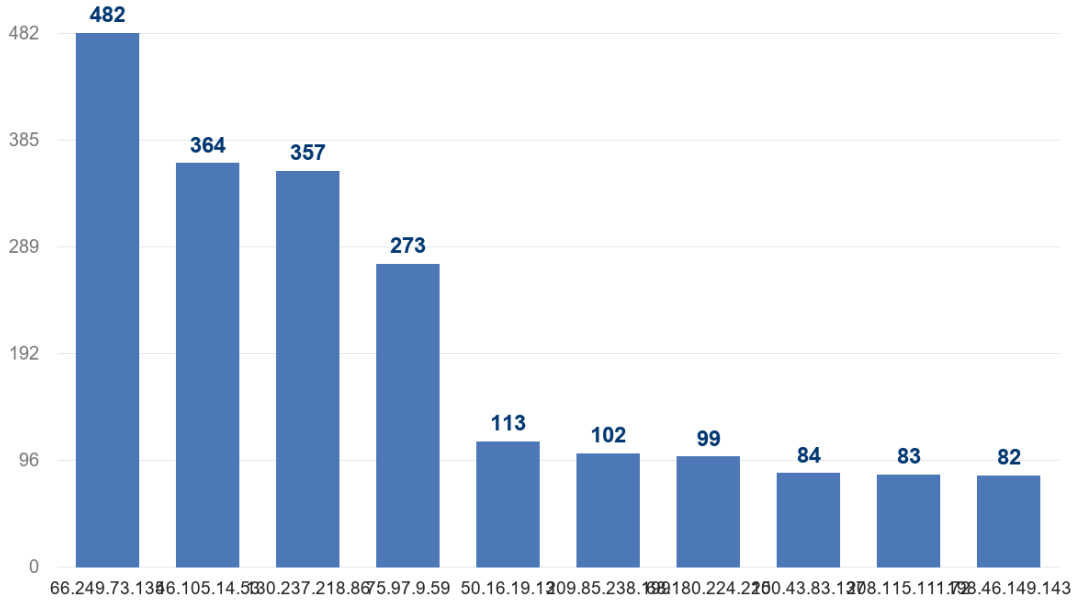
Metric	Value
Total lines / requests parsed	10,000
Unique client IP addresses	1,753
Distinct source-to-destination flows	5 IP pairs in the packet capture, thousands of client-server pairs here
Methods used	GET 9,952, HEAD 42, POST 5, OPTIONS 1
Successful responses (2xx)	9,171
Redirects (3xx)	609
Client errors (4xx)	217
Server errors (5xx)	3

3. The answer the task asks for - the top 3 IP addresses

Rank	IP address	Requests	Share of the log	Character of the traffic
1	66.249.73.135	482	4.82%	Consistent, polite, crawler-style access (Googlebot range)
2	46.105.14.53	364	3.64%	Feed reader - the `UniversalFeedParser` user-agent appears exactly 364 times
3	130.237.218.86	357	3.57%	Highest-volume interactive client, mixed assets and pages

Apache Access Log - Top 10 Client IP Addresses

Requests per client IP address (bar label = number of requests)



Source: real Apache combined access log (10,000 requests, 17-22 May 2015)

Task 4 - Top 3 Source IP Addresses in the Apache Access Log

REAL DATA

10,000 real requests in Apache 'combined' format (17-20 May 2015)

Rank	Client IP address	Requests	Share of all traffic	Note
1	66.249.73.135	482	4.82%	Consistent crawler-style access
2	46.105.14.53	364	3.64%	Feed reader (UniversalFeedParser user-agent)
3	130.237.218.86	357	3.57%	Highest-volume interactive client

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Commands that produce exactly this table:

```
cut -d ' ' -f 1 apache_logs | sort | uniq -c | sort -nr | head -3
awk '{print $1}' apache_logs | sort | uniq -c | sort -rn | head -3
```

How to read the result: together the top three addresses account for about 12% of all traffic - normal for a public website, because crawlers and feed readers poll the same content repeatedly. If a single address were responsible for 40-60% of requests, that would be the moment to look closer.

4. The other counts, and why they matter

Status codes

Code	Meaning	Count	Analyst comment
200	OK	9,126	Normal serving
206	Partial content	45	Ranged downloads (video/audio seeking)
301	Moved permanently	164	Legitimate redirects
304	Not modified	445	Caching working as designed
404	Not found	213	Resource discovery / bad links
403	Forbidden	2	Access control refusing
416	Range not satisfiable	2	Client asked for an invalid byte range
500	Internal server error	3	Application faults worth checking

Task 4 - Status Code Distribution

REAL DATA

How the server answered the 10,000 requests (awk '$9 == 200' | wc -l style analysis)

Status code	Meaning	Requests
200		9126
206		45
301		164
304		445
403		2
404		213
416		2
500		3
TOTAL	2xx=9171 3xx=609 4xx=217 5xx=3	10000

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The most suspicious finding in the whole log

The 404s are not random. They cluster on content-management-system admin paths, which is the signature of automated discovery:

Client IP	Errors	URLs requested	Hits	Interpretation
208.91.156.11	60	`/wp-login.php`, `/wp-login.php?action=register`, `/wp-admin/`, `/wp-content/`	6 each	WordPress login/page probing
144.76.95.39	14	`/blog/wp-admin/`, `/wordpress/wp-admin/`	6	WordPress admin discovery
66.249.73.135	10	`/wordpress/wp-admin/`, `/wp-admin/`	5	Crawler following old links

Client IP	Errors	URLs requested	Hits	Interpretation
91.236.75.25	8	`/administrator/index.php`	2	Joomla admin probing - a different CMS than the site runs
75.97.9.59	6	`/admin.php`	4	Generic admin-page probing

Task 4 - Suspicious Requests Found During the Review

REAL DATA

Scanner-style URLs and the IPs that generated the most client errors

Client IP	Errors	URL requested	Hits	Interpretation
208.91.156.11	60	/wp-login.php, /wp-admin/	6	WordPress login probing
144.76.95.39	14	/blog/wp-admin/	6	WordPress admin discovery
91.236.75.25	8	/administrator/index.php	2	Joomla admin probing
66.249.73.135	10	/wordpress/wp-admin/	5	Crawler hitting admin paths
75.97.9.59	6	/admin.php	4	Generic admin page probing

Automated content-management-system discovery - the classic low-and-slow web scan pattern

Why the Joomla path matters most: the server is not a Joomla site, so a request for `/administrator/index.php` cannot be a user typing a URL - it is a scanner running a word list. That is a single-request proof of hostile intent, and it is exactly the kind of evidence a SOC analyst escalates.

Recommended follow-up actions, in order:

1. Block the top offenders at the WAF/firewall (`208.91.156.11`, `91.236.75.25`, `144.76.95.39`).
2. Confirm the site does not actually expose `/wp-login.php` or `/administrator/`.
3. Add a detection rule: more than N 404s from one IP in 60 seconds is a scanning alert.
4. Check whether the 3 × HTTP 500 responses line up in time with the scanner activity.
5. Review the 5 POST requests individually - POSTs are rare in this log, so each one deserves a look.

User agents

User-Agent	Requests
`Mozilla/5.0 (Windows NT 6.1; WOW64) ... Chrome/32.0.1700.107`	1,044
`Mozilla/5.0 (Macintosh; ...) Chrome/33.0.1750.91`	369
`UniversalFeedParser/4.2-pre-314-svn +http://feedparser.org/`	364
`Mozilla/5.0 (Windows NT 6.1; WOW64; rv:27.0) Gecko/20100101 Firefox/27.0`	296
`Mozilla/5.0 (iPhone; CPU iPhone OS 6_0 ...) Version/6.0 Mobile`	271

The third entry is a library name, not a browser - which is how IP `46.105.14.53` was identified as a feed reader without looking at a single URL.

5. ATT&CK mapping for what was found

Behaviour	Tactic	Technique
Requesting a list of CMS admin and login paths	Reconnaissance	T1595.002 - Active Scanning: Vulnerability Scanning

Behaviour	Tactic	Technique
Repeated 404s from one source in a short window	Reconnaissance	T1595 - Active Scanning
Probing `/wp-login.php` and `/administrator/index.php` for a login form	Initial Access (if used)	T1190 - Exploit Public-Facing Application (attempt stage)
Crawler/feed-reader traffic at scale	-	Benign, useful as a baseline

6. Tools used

Step	Tool
Read the raw file	Notepad++ / VS Code (any text editor)
Count with the shell	`cut`, `awk`, `sort`, `uniq`, `grep`, `wc`
Verify in a spreadsheet	Microsoft Excel - open as CSV, filter column I for `404`, pivot on column A
Cross-check at scale	A short Python script (the figures in this document were produced and verified this way)

7. Screenshots to take

1. The log open in a text editor with line numbers and the first few requests visible.
2. Excel/Notepad++ with the `uniq -c` output or a pivot table showing the top 3 IP addresses - save as `4e Top 3 IP counts in the editor.png`.
3. The command-line pipeline and its output in a terminal - save as `4f Command line counting.png`.